Assignment -02

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Subject – DSA Recursion

Q1. Print Numbers from 1 to N

```java
import java.util.Scanner;
public class Number1ton {  new *

    static void printIncrease(int n) {  2 usages  new *
        if (n == 1) {
            System.out.println(1);
            return;
        }
        printIncrease( n: n-1);
        System.out.println(n);
    }
    public static void main(String[] args) {  new *
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        printIncrease(n);

    }
}
```

Q2. Print Numbers from N to 1

```java
import java.util.Scanner;
public class NumberNto1 {  new *
    static void printDecrease(int n){  2 usages  new *
        if(n==1){
            System.out.println(1);
            return;
        }
        System.out.println(n);
        printDecrease( n: n-1);
    }
    public static void main(String[] args){  new *
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        printDecrease(n);
    }
}
```

Q3. Print Even Numbers from 1 to N

```java
import java.util.Scanner;
public class EvenANDOdd {  new *
    static void even(int n) {  2 usages  new *
        if (n <= 0) {
            return;
        }
        even( n: n - 1);
        if (n % 2 == 0) { //n%!=2==0 for odd
            System.out.println(n);
        }
    }
    public static void main(String[] ages){  new *
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        even( n: 5);
    }
}
```

Q4. Print Odd Numbers from 1 to N

```java
import java.util.Scanner;
public class EvenANDOdd {  new *
    static void even(int n) {  2 usages  new *
        if (n <= 0) {
            return;
        }
        even( n: n - 1);
        if (n % 2 != 0) { //n%=2!=0 for odd
            System.out.println(n);
        }
    }
    public static void main(String[] ages){  new *
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        even( n: 5);
    }
}
```

Q5. Sum of Digit N Natural Numbers

```java
import java.util.Scanner;
public class SumOfDigit {  new *
    static int sumOfdigit(int n){  2 usages  new *
        if(n==0){
            return 0;
        }
        int sum = n%10+sumOfdigit( n: n/10);
        return sum;
    }
    static void main(String[] args) {  new *
        Scanner sc = new Scanner(System.in);
        int n =sc.nextInt();
        int ans = n%10+sumOfdigit( n: n/10);
        System.out.println(ans);
    }
}
```

Q6. Factorial of a Number

```java
import java.util.Scanner;
public class SumOfDigit {  new *
    static int sumOfdigit(int n){  2 usages  new *
        if(n==0){
            return 0;
        }
        int sum = n%10+sumOfdigit( n: n/10);
        return sum;
    }
    static void main(String[] args) {  new *
        Scanner sc = new Scanner(System.in);
        int n =sc.nextInt();
        int ans = n%10+sumOfdigit( n: n/10);
        System.out.println(ans);
    }
}
```

Q7. Reverse a String

```java
import java.util.Scanner;
public class ReverseString {  new *
    static void rev(String s , int i){  2 usages  new *
        if(i<0){
            return;
        }
        System.out.print(s.charAt(i));
        rev(s, i: i-1);
    }
    public static void main(String[] args){  new *
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
        rev(s, i: s.length()-1);
    }
}
```

Q8. Check Whether a String Is a Palindrome

```java
import java.util.Scanner;
Rename usages
public class CheckPalindrome{  new *
    static boolean helper(String str , int s, int e){  2 usages  new *
        if(s>=e){
            return true;
        }
        if(str.charAt(s)!= str.charAt(e)){
            return false;

        }
        return helper(str,  s: s+1, e: e-1);
    }
    public static void main(String[] args){  new *
        Scanner sc = new Scanner(System.in);
        String str = sc.nextLine();
        helper(str, s: 0, e: str.length()-1);
    }
}
```

Q9. Reverse a Number

```java
import java.util.Scanner;
public class ReverseNumber {  new *
    static int reverse(int n , int rev ){  2 usages  new *
        if(n==0){
            return rev;
        }
        return reverse( n: n/10, rev: rev * 10 + n%10);

    }
    public static void main(String[] args){  new *
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int ans = reverse(n, rev: 0);
        System.out.println("result:"+ ans);
    }
}
```

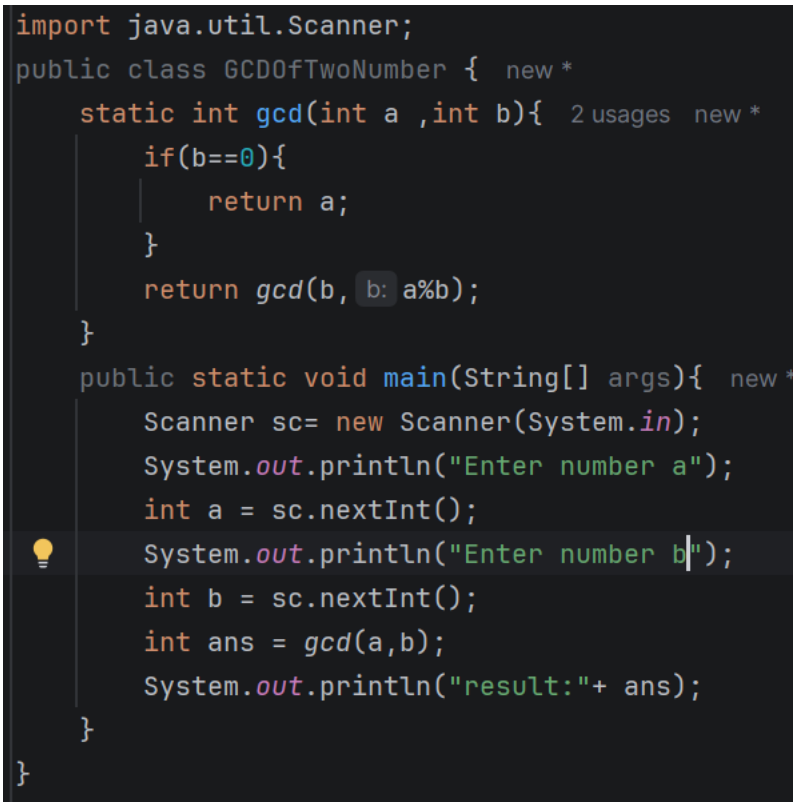
Q13. Power of a Number

```java
import java.util.Scanner;
public class aPowerb {  new *
    static int power(int a, int b){  2 usages  new *
        if(b==0){
            return 1;
        }
        int power = a * power(a, b: b-1);
        return power;
    }
    static void main(String[] args) {  new *
        Scanner sc = new Scanner(System.in);
        int a  = sc.nextInt();
        int b = sc.nextInt();
        int ans = a*power(a, b: b-1);
        System.out.println(ans);
    }
}
```

Q14. Fibonacci Series

```java
import java.util.Scanner;
public class FibonacciNumber {  new *
    static  int   fib(int n){  3 usages  new *
        if(n==0 || n==1){
            return n;
        }
        int previous = fib( n: n-1);
        int previous_previous = fib( n: n-2);
        return previous+previous_previous;
    }
    public static void main(String[] args){  new *
        Scanner sc = new Scanner(System.in);
        System.out.println(fib( n: 7));
    }
}
```

Q15. GCD of Two Numbers



```java
import java.util.Scanner;
public class GCDOfTwoNumber {  new *
    static int gcd(int a ,int b){  2 usages  new *
        if(b==0){
            return a;
        }
        return gcd(b, b: a%b);
    }
    public static void main(String[] args){  new *
        Scanner sc= new Scanner(System.in);
        System.out.println("Enter number a");
        int a = sc.nextInt();
        System.out.println("Enter number b");
        int b = sc.nextInt();
        int ans = gcd(a,b);
        System.out.println("result:"+ ans);
    }
}
```

Q22. Check Whether an Array Is Sorted

```java
import java.util.Scanner;
public class CheckArraySort {  new *
    static boolean sorted(int []arr ,int index){  2 usages  new
        if(index >= arr.length-1){
            return true;
        }
        if(arr[index]> arr[index+1]){
            return false;
        }
        return sorted(arr, index: index+1);
    }
    public static void main(String[] args){  new *
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int [] arr = new int[n];
        for(int i =0; i<n; i++){
            arr[i] = sc.nextInt();

        }
        if(sorted(arr, index: 0)){
            System.out.println("Array is sorted");
        }else{
            System.out.println("Array is not sorted");
        }
    }
```

Q23. Find the Maximum Element in an Array

```java
import java.util.Scanner;
public class MaximumElement {  new *

    public static void main(String[] args) {  new *
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter array size: ");
        int n = sc.nextInt();

        if (n <= 0) {
            System.out.println("Array size must be positive");
            sc.close();
            return;
        }

        int[] arr = new int[n];

        System.out.println("Enter array elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int max = arr[0];

        for (int i = 1; i < n; i++) {
            if (arr[i] > max) {
                max = arr[i];
            }
        }

        System.out.println("Maximum element = " + max);

        sc.close();
    }
}
```

Q24. Find the Minimum Element in an Array

```java
import java.util.Scanner;
public class MinimumElement {  new *
    public static void main(String[] args) {  new *
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter array size: ");
        int n = sc.nextInt();
        if (n <= 0) {
            System.out.println("Array size must be positive");
            sc.close();
            return;
        }
        int[] arr = new int[n];

        System.out.println("Enter array elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int min = arr[0];

        for (int i = 1; i < n; i++) {
            if (arr[i] < min) {
                min = arr[i];
            }
        }

        System.out.println("Minimum element = " + min);

        sc.close();
    }
}
```

Q25. Binary Search Using Recursion

```java
import java.util.Scanner;

public class BinarySearch {  new *
    static int binary(int []arr, int s, int e , int target){  3 usages  new *
        if(s>e){
            return -1;
        }
        int mid = s+(e-s)/2;
        if(arr[mid] == target){
            return mid;
        }
        if(arr[mid]> target){
           return binary(arr , s, e: mid-1,target);
        }
        return binary(arr, s: mid+1, e, target);
    }
    public static void main(String[] args){  new *
        Scanner sc = new Scanner(System.in);
        int n= sc.nextInt();
        int []arr = new int[n];
        for(int i =0; i<n; i++){
            arr[i] = sc.nextInt();
        }
        int target = sc.nextInt();
        int ans = binary(arr,  s: 0 , e: n-1, target);
        if(ans == -1){
            System.out.println("Element is not found");
        }else{
            System.out.println("Element is found:" + ans);
        }
    }
}
```

Q26. Linear Search Using Recursion

```java
import java.util.Scanner;
public class LinearSearch {  new *
    static int search(int[] arr, int target, int index) {  2 usages  new *
        if (index == arr.length) {
            return -1;
        }
        if (arr[index] == target) {
            return index;
        }
        return search(arr, target,  index: index + 1);
    }
    public static void main(String[] args) {  new *
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter array size: ");
        int n = sc.nextInt();

        int[] arr = new int[n];
        System.out.println("Enter array elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        System.out.print("Enter element to search: ");
        int target = sc.nextInt();

        int result = search(arr, target,  index: 0);

        if (result == -1) {
            System.out.println("Element not found");
        } else {
            System.out.println("Element found at index: " + result);
        }
    }
}
```